

Git Hub Hands on Learning -4

Creating a new branch name GitWork

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

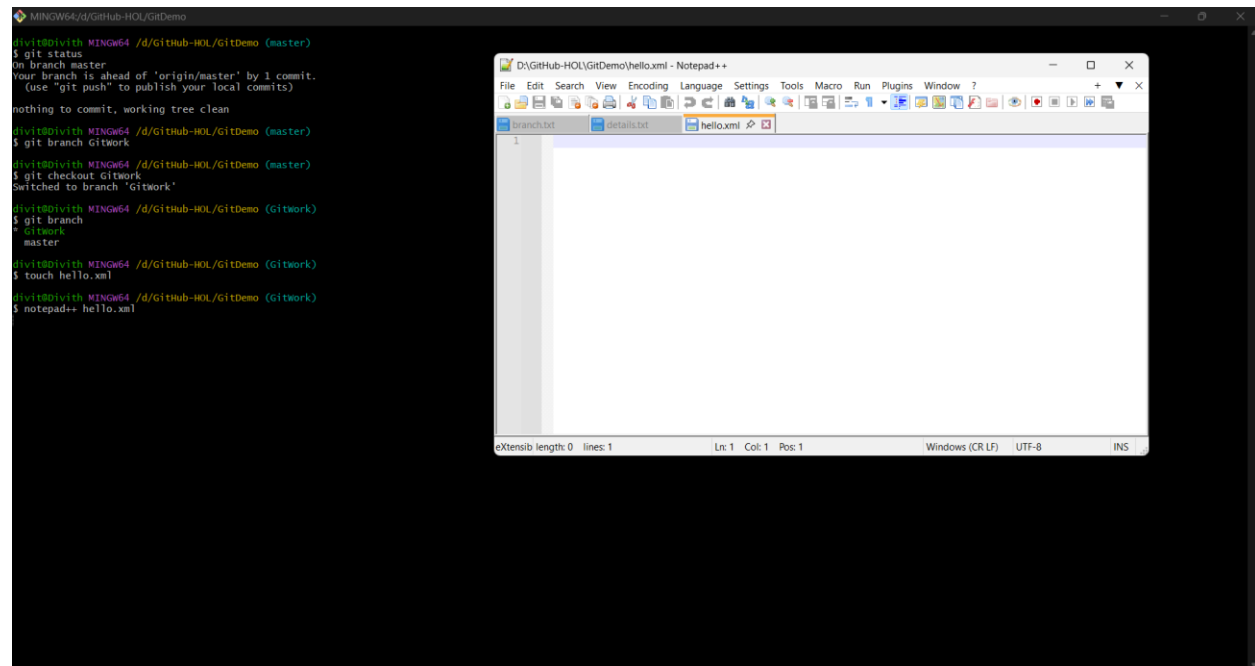
nothing to commit, working tree clean

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch GitWork

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git checkout GitWork
Switched to branch 'GitWork'

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitWork)
$ git branch
* GitWork
  master
```

Creating a new file touch.xml and modifying it at GitWork Branch



Adding to Commit Process

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitWork)
$ git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitWork)
$ git add hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitWork)
$ git commit -m "Added hello.xml in GitWork"
[GitWork 9cec3bc] Added hello.xml in GitWork
1 file changed, 5 insertions(+)
create mode 100644 hello.xml

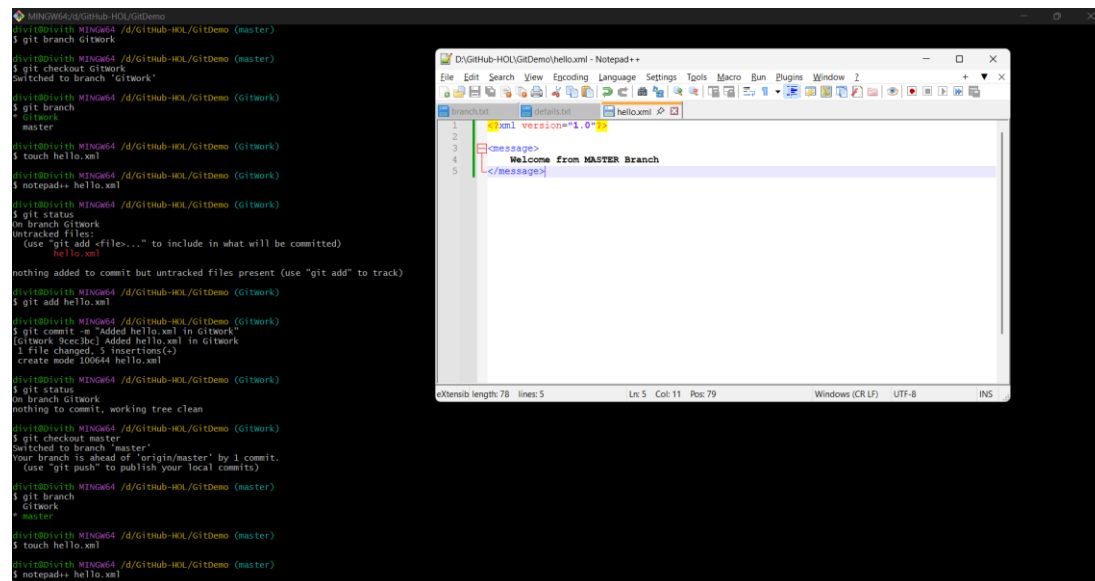
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitWork)
$ git status
On branch GitWork
nothing to commit, working tree clean
```

Switching to Master Branch

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (GitWork)
$ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch
  GitWork
* master
```

Adding touch.xml and modifying it in Master Branch



The screenshot shows a terminal window on the left and a Notepad++ editor on the right. The terminal window displays the following commands and output:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch GitWork
$ git checkout GitWork
Switched to branch 'GitWork'
$ git branch
  GitWork
* master
$ touch hello.xml
$ notepad++ hello.xml
$ git status
On branch GitWork
Untracked files:
  (use "git add <file>..." to include in what will be committed)
    hello.xml

nothing added to commit but untracked files present (use "git add" to track)
$ git add hello.xml
$ git commit -m "Added hello.xml in GitWork"
[GitWork 9cec3bc] Added hello.xml in GitWork
1 file changed, 5 insertions(+)
create mode 100644 hello.xml
$ git status
On branch GitWork
nothing to commit, working tree clean
$ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)
$ git branch
  GitWork
* master
$ touch hello.xml
$ notepad++ hello.xml
```

The Notepad++ editor on the right shows the content of the file `hello.xml`, which is:

```
<?xml version="1.0"?>
<message>
  Welcome From MASTER Branch
</message>
```

Adding to Commit Process:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git add hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git commit -m "Added hello.xml in Master"
[master e1c3df4] Added hello.xml in Master
1 file changed, 5 insertions(+)
create mode 100644 hello.xml
```

Log Status:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git log --oneline --graph --decorate --all
* e1c3df4 (HEAD -> master) Added hello.xml in Master
| * 9cec3bc (GitWork) Added hello.xml in GitWork
|/
* 39ac866 Added files in GitNewBranch
* f6a6bcd (origin/master) Added gitignore to ignore log files
* a7fa51c Added welcome.txt file
```

Difference between Master and GitWork Branch:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git diff master GitWork
diff --git a/hello.xml b/hello.xml
index a84a518..71f1eec 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,5 +1,5 @@
 <?xml version="1.0"?>

 <message>
-  Welcome from MASTER Branch
+  Welcome from GitWork Branch
 </message>
\ No newline at end of file
```

Git Merge Failing due to Conflict of hello.xml file:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.
```

Conflict Markup Output:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo
$ git checkout master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
(use "git push" to publish your local commits)

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch
* master

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ touch hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ notepad++ hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git add hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git commit -m "Added hello.xml in Master"
[master e1c3df4] Added hello.xml in Master
1 file changed, 5 insertions(+)
create mode 100644 hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git log --oneline --graph --decorate --all
* e1c3df4 (HEAD -> master) Added hello.xml in Master
* 9cec3bc (GitWork) Added hello.xml in GitWork
/
* 39ac866 Added files in GitNewBranch
* f6ab6cd (origin/master) Added gitignore to ignore log files
* a7ab1c Added welcome.txt file

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git diff master GitWork
diff --git a/hello.xml b/hello.xml
index a84a518..71f1ee0 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,5 +1,5 @@
<?xml version="1.0"?>

<message>
- welcome from MASTER Branch
+ welcome from GitWork Branch
</message>
\ No newline at end of file

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master|MERGING)
$ notepad++ hello.xml
```

Edited the File manually.

Merge Resolving:

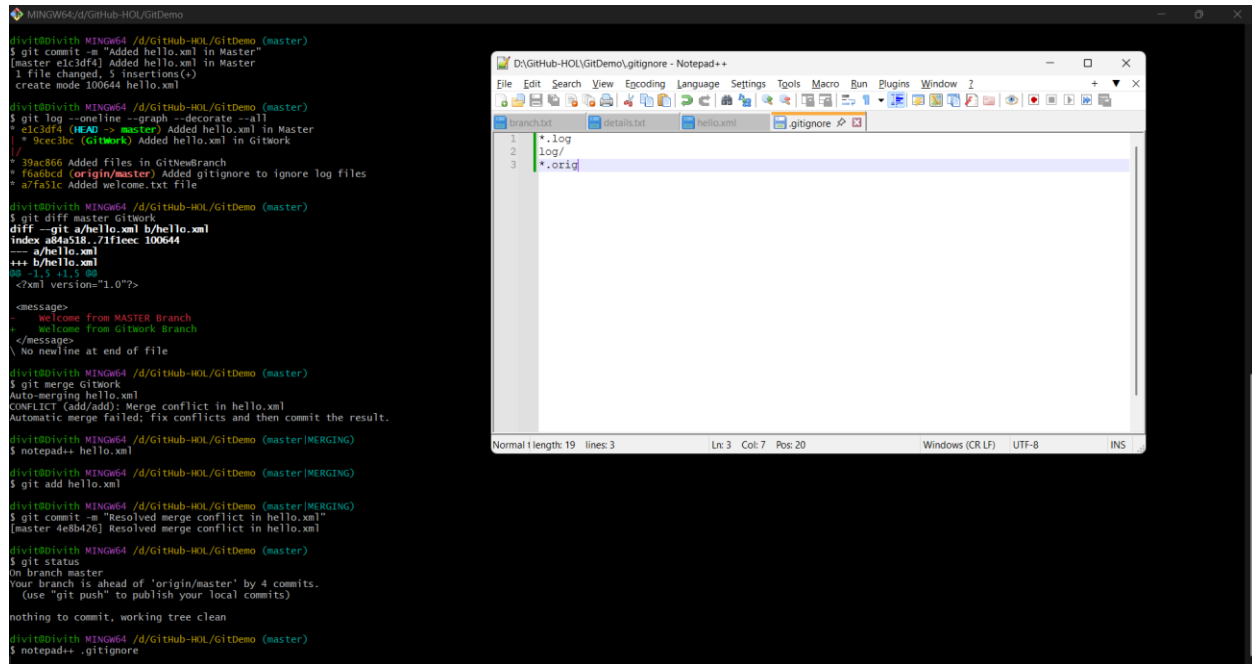
```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master|MERGING)
$ git add hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master|MERGING)
$ git commit -m "Resolved merge conflict in hello.xml"
[master 4e8b426] Resolved merge conflict in hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 4 commits.
(use "git push" to publish your local commits)

nothing to commit, working tree clean
```

Updating the .gitignore:



The screenshot shows a terminal window on the left and a Notepad++ editor on the right. The terminal window displays the following commands and output:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git commit -m "Added hello.xml in Master"
[master e1c3df4] Added hello.xml in Master
1 file changed, 1 insertion(+)
create mode 100644 hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git log --oneline --graph --decorate --all
* e1c3df4 (HEAD -> master) Added hello.xml in Master
/* 9cec3bc (GitWork) Added hello.xml in GitWork
/* 39ac866 Added files in GitNewBranch
/* f6ab6cd (origin/master) Added gitignore to ignore log files
/* a7fa51c Added welcome.txt file

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git diff master GitWork
diff --git a/hello.xml b/hello.xml
index a84a518..7f1fec 100644
--- a/hello.xml
+++ b/hello.xml
@@ -1,3 +1,4 @@
<?xml version="1.0"?>

<message>
- welcome from MASTER Branch
+ welcome from GitWork Branch
</message>
\ No newline at end of file

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git merge GitWork
Auto-merging hello.xml
CONFLICT (add/add): Merge conflict in hello.xml
Automatic merge failed; fix conflicts and then commit the result.

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master/MERGING)
$ notepad++ hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master/MERGING)
$ git add hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master/MERGING)
$ git commit -m "Resolved merge conflict in hello.xml"
[master 4eb426] Resolved merge conflict in hello.xml

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git status
On branch master
Your branch is ahead of 'origin/master' by 4 commits.
(Use "git push" to publish your local commits)

nothing to commit, working tree clean

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ notepad++ .gitignore
```

The Notepad++ editor shows the content of the .gitignore file:

```
1 *.log
2 log/
3 *.orig
```

Commit of .gitignore files

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git add .gitignore

divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git commit -m "Updated gitignore for backup files"
[master 328e60c] Updated gitignore for backup files
1 file changed, 2 insertions(+), 1 deletion(-)
```

Checking the Branch:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch
GitWork
* master
```

Delete the Branch:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git branch -d GitWork
Deleted branch GitWork (was 9cec3bc).
```

Log View:

```
divit@Divith MINGW64 /d/GitHub-HOL/GitDemo (master)
$ git log --oneline --graph --decorate
* 328e60c (HEAD -> master) Updated gitignore for backup files
*   4e8b426 Resolved merge conflict in hello.xml
| \
|  * 9cec3bc Added hello.xml in GitWork
* | e1c3df4 Added hello.xml in Master
| /
* 39ac866 Added files in GitNewBranch
* f6a6bcd (origin/master) Added gitignore to ignore log files
* a7fa51c Added welcome.txt file
```